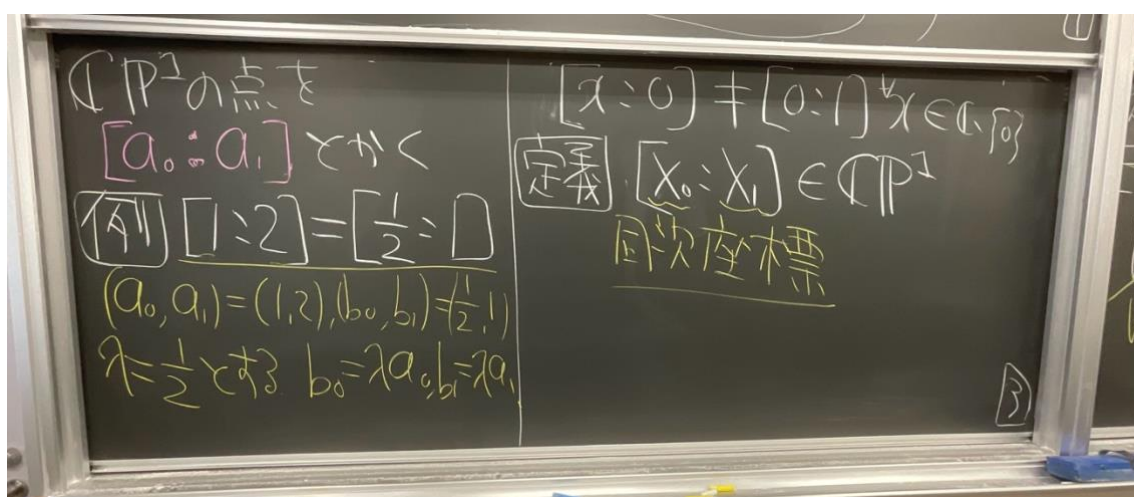
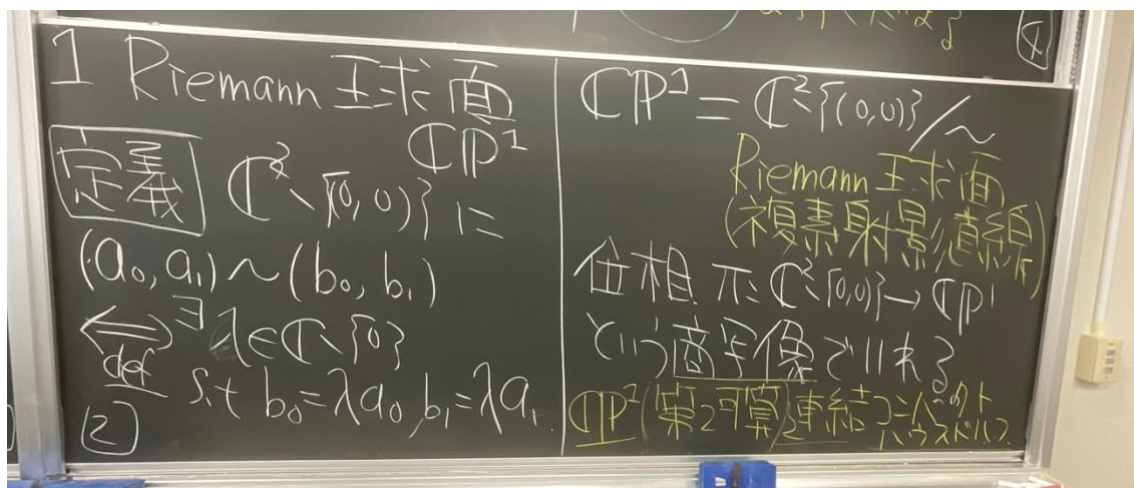
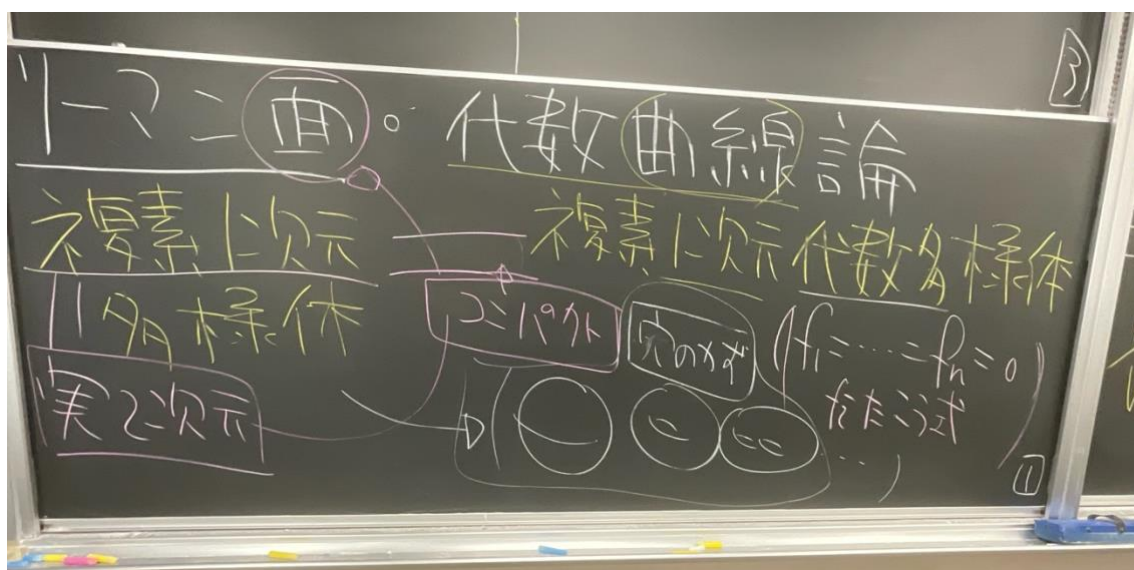




$$[A] + (x_0, x_1) = x_0^4 - x_1^4$$

$$F: \mathbb{CP}^1 \rightarrow \mathbb{C}$$

$$[a_0: a_1] \rightarrow F(a_0, a_1)$$

Well-defined?

$$[2:1] = [1:\frac{1}{2}] \text{ なのに } F(2,1) \neq F(1,\frac{1}{2}) \in \mathbb{C}$$

$$V(F) = \{ [a_0: a_1] \in \mathbb{CP}^1 \mid a_0^4 - a_1^4 = 0 \}$$

(a_0, a_1) の $\frac{1}{4}$ 倍は $\frac{1}{4}$ 倍

$$U_0 = \{ [a_0: a_1] \in \mathbb{CP}^1 \mid a_0 \neq 0 \}$$

$$U_1 = \{ [a_0: a_1] \in \mathbb{CP}^1 \mid a_1 \neq 0 \}$$

$$\mathbb{CP}^1 = U_0 \cup U_1$$

$$U_0 \cap U_1 = U_0 \setminus \{ [1:0] \}$$

$$= U_1 \setminus \{ [0:1] \}$$

補題 1.6

$$\varphi_0: U_0 \rightarrow \mathbb{C}_w$$

$$[a_0: a_1] \rightarrow \frac{a_1}{a_0}$$

$$\varphi_0^{-1}: \mathbb{C}_w \rightarrow U_0$$

$$w \mapsto [1:w]$$

同相

$[a_0: a_1] = [\frac{a_0}{a_1}: 1]$

$$\varphi_1: U_1 \rightarrow \mathbb{C}_z$$

$$[a_0: a_1] \rightarrow \frac{a_0}{a_1}$$

$$\varphi_1^{-1}: \mathbb{C}_z \rightarrow U_1$$

$$z \mapsto [z: 1]$$

$\tilde{S}^1 =$

$$\varphi_1^{-1}: (\mathbb{C}_z \setminus \{0\}) \rightarrow (\mathbb{C}_w \setminus \{0\})$$

$$z \mapsto 1/z$$

$\varphi_0 \circ \varphi_1^{-1}(z) = \varphi_0([z: 1]) = \frac{1}{z}$

$\mathbb{C}_w$

$\mathbb{C}_z$

$\mathbb{CP}^1$

$\tilde{S}^1$

$\mathbb{CP}^1 \subset \mathbb{C}P^1$ は同相

$$\begin{array}{|l}
 \text{定義 } f: \mathbb{CP}^1 \rightarrow \mathbb{C} \\
 f_z = f \circ \varphi_1^{-1}(z) = f\left(\frac{1}{z}\right) \\
 f_w = f \circ \varphi_0^{-1}(w) = f(1-w) \\
 \end{array}
 \quad
 \begin{array}{|l}
 \text{例 } f: \mathbb{CP}^1 \rightarrow \mathbb{C} \\
 [a_0: a_1] \mapsto \frac{a_0 \bar{a}_1}{|a_0|^2 + |a_1|^2} \\
 f_z = f\left(\frac{1}{z}\right) = \frac{z}{|z|^2 + 1} \\
 f_w = f(1-w) = \frac{w}{1 + |w|^2} \quad \text{③}
 \end{array}$$

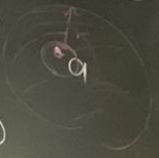
$$\begin{array}{|l}
 \text{例 } f_z\left(\frac{1}{w}\right) = \frac{\frac{1}{w}}{\left|\frac{1}{w}\right|^2 + 1} \\
 = \frac{w}{1 + |w|^2} \\
 = f_w(w) \\
 \text{逆写}
 \end{array}
 \quad
 \begin{array}{|l}
 \text{補題 } f_z, f_w: \mathbb{C} \rightarrow \mathbb{C} \\
 f_z\left(\frac{1}{w}\right) = f_w(w) \quad w \neq 0 \\
 \exists f: \mathbb{CP}^1 \rightarrow \mathbb{C} \\
 \text{s.t. } f_z = f \circ \varphi_1^{-1}, f_w = f \circ \varphi_0^{-1} \quad \text{④}
 \end{array}$$

$$\begin{array}{|l}
 1.2 \text{ 複素解析} \\
 f: U \rightarrow \mathbb{C} \text{ 正則} \\
 \Leftrightarrow \forall z \in U \quad (U \subset \mathbb{C} \text{ domain}) \\
 \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} \text{ 存在} \\
 (\text{Cauchy-Riemann})
 \end{array}
 \quad
 \begin{array}{|l}
 \text{正則} \Leftrightarrow \forall a \in U, \exists r > 0 \\
 f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z-a)^n \\
 \text{①}
 \end{array}$$

定理 $f, g: U \rightarrow \mathbb{C}$
 $f|_V = g|_V \Rightarrow f = g$ (正則)
 $\exists V \subset U$
 V open
Rouville の定理
 $f: D \rightarrow \mathbb{C}$ 正則 + 有界 $\Rightarrow$ 定数

定義 Laurent 展開
 $a \in \mathbb{C}, f: 0 < |z-a| < r$ 上正則
 $\Rightarrow f(z) = \sum_{n=-\infty}^{\infty} C_n (z-a)^n$

① $C_{-1} = C_{-2} = \dots = 0$
 $\Rightarrow$ 除去可能特異点
 $(f \text{ が正則になる } (z-a) \text{ 点})$



② $C_{-1} = C_{-2} = \dots = 0$
重要 木取 $f(z) = \frac{h(z)}{(z-a)^m}$
 h 正則
 ③ それ以外
真性特異点
(位数, a は m 位の極)

有理型関数 $(a, a_n \text{ 有限個})$
 $a_1, a_2, \dots \in U$ 集積点がない
 $f: U \setminus \{a_1, a_2, \dots\} \rightarrow \mathbb{C}$ 正則
 a_i が ① or ② のとき
 f を有理型関数

例 $f(z) = \frac{1+z}{z^2}$
 $= \frac{(1+z)}{(z-0)^2}$
 0 は 2 位の極

1.3 $\mathbb{CP}^1$ 上の正則関数
定義 $V \subset \mathbb{CP}^1$ 令 V 正則
 $(f: V \rightarrow \mathbb{C})$
 f_z, f_w (前に z, w したものとす)
 ① f が正則 $\Leftrightarrow f_z, f_w$ 正則
 ② f が有理型 $\Leftrightarrow$ 有理型

$f: \mathbb{CP}^1 \rightarrow \mathbb{C}$ $[a: a_1] \rightarrow 1$ 正則 $f_z = f \circ \varphi_1^{-1}(z) = f([z: 1])$ $f_w = 1$	$f: \mathbb{CP}^1 \rightarrow \mathbb{C}$ $[a_0: a_1] \rightarrow \frac{a_0}{a_1}$ 有理型 $f_z = f([z: 1]) = z$ 正則 $f_w = f([1: w]) = \frac{1}{w}$ not 正則 有理型 (1位の極)
--	--

定理

$\mathbb{CP}^1$ 上の正則関数は定数のみ
